

GEOMETRY

2026 Spring

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Chapter 0

Introduction

I was preparing Yau's contest and Qiuzhen qualifying exams in 2026 spring, and wrote this problem set about geometry.

The problem set mainly contains exercise on Lee's book, Qiuzhen qualifying exams, and Yau's contest. I select some interesting and challenging exercise to write, and the ones which is easy to write answers(hhh). Since the time is limited, this problem set contains still a little exercise. And I will update from time to time, and add more exercise, on my github ' <https://tianyc089.github.io/> '.

The answers must have some typos or mistakes, or if you have questions on my answers, or you have some fantastic problem want to tell me, welcome to contact me in email. I will appreciate a lot.

Chapter 1

Differential geometry

1.1 Exercise

Exercise 1.1.1 ([3] 25 spring T1). *Let M, N be closed, connected manifolds of **the same dimension**. Suppose $f : M \rightarrow N$ is an immersion. Is the following true or false? If true, prove your claim. If false, give a counterexample and prove that your example is indeed a counterexample.*

- (a) f is injective.
- (b) f is surjective.
- (c) f is a homeomorphism.
- (d) f is a covering map.

Proof. (a)(c) are false since we may take $f : S^1 \rightarrow S^1, z \mapsto z^2$. Now let's prove (b)(d):

- (b) Since f an immersion, df_p is injective for any $p \in M$. But $\dim M = \dim N$, hence df_p is an isomorphism, thus f is a local diffeomorphism. Now we want to show that $f(M)$ is both open and closed in N , hence $f(M) = N$ since N is connected.

For any $q = f(p)$, since f is a local diffeomorphism, there exists neighborhoods V of q and U of p such that $f|_U : U \rightarrow V$ is a diffeomorphism. Thus $V \in f(M)$, that is, $f(M)$ is open. On the other hand, since M is compact, $f(M)$ is compact. Note that N is Hausdorff, hence $f(M)$ is closed.

- (d) We have proved that f is surjective. Now we want to prove the covering condition. For any $q \in N$, since $\{q\}$ is closed in N , $f^{-1}(q)$ is closed in M . But at any $p \in f^{-1}(q)$, f is local diffeomorphism, thus $f^{-1}(q)$ consists only finitely many points. Otherwise, there exists a limit point $p_0 \in f^{-1}(q)$, local diffeomorphism at p_0 gives a contradiction! Finally, we can take a small neighborhood V of q , then $f^{-1}(V) = \bigsqcup_{i=1}^k U_i$, $f|_{U_i}$ is a diffeomorphism for each i .

□

Chapter 2

Riemannian metric, geodesics

2.1 Exercise

Exercise 2.1.1 ([3] 26 spring T8). Let $\Omega = \mathbb{R}^n \setminus \{0\}$ and $u(r) = 1 + r^{2-n}$ (with $r = |x|$), $n \geq 3$. Consider the metric $g_u = u^{\frac{4}{n-2}} \delta$, where δ is the Euclidean metric. Decide whether (Ω, g_u) is complete, and justify your answer.

Proof. By Hopf-Rinow theorem, we only need to check whether geodesics can be extended infinitely at a single point. We may assume the point is $x_0 = (1, 0, 0, \dots, 0)$. Since g is spherically symmetric, we may examine the radial geodesics.

Note that $ds^2 = u^{\frac{4}{n-2}} \delta$, hence $ds = u^{\frac{2}{n-2}} dr$. Hence the length of the radial geodesic starting at x_0 to infinity is

$$\int_1^\infty (1 + r^{2-n})^{\frac{2}{n-2}} dr \sim \int_1^\infty 1 dr$$

which is divergent. Hence it can be extended infinitely to infinity. The length of the radial geodesic starting at x_0 to 0 is

$$\int_0^1 (1 + r^{2-n})^{\frac{2}{n-2}} dr \sim \int_0^1 r^{-2} dr$$

which is also divergent. Hence it can be extended infinitely to 0. Thus (Ω, g_u) is complete. $\square$

Exercise 2.1.2. For which $a \in \mathbb{R}$, $(\mathbb{R}^2, g_a)$ is complete, where $g_a = e^{a(x^2+y^2)}(dx^2 + dy^2)$.

Proof. As the same in last exercise, we only need to consider

$$\int_0^\infty e^{\frac{a}{2}r^2} dr$$

The integral diverges if and only if $a \geq 0$. Hence $(\mathbb{R}^2, g_u)$ is complete if and only if $a \geq 0$. $\square$

Exercise 2.1.3 ([3] 25 autumn T6). Let (M, g) be a Riemannian manifold and $\pi : \widetilde{M} \rightarrow M$ be a covering map with the pullback metric π^*g on $\widetilde{M}$.

(a) Show that M is complete if and only if $\widetilde{M}$ is complete.

(b) Is the conclusion in (a) true if π is only assumed to be a local isometry? If so, provide a proof. If not, provide a counterexample.

Proof. (a) First suppose $\widetilde{M}$ is complete. For any $p \in M$ and $v \in T_p M$, let $\tilde{p} \in \widetilde{M}$ and $\tilde{v} \in T_{\tilde{p}}(\widetilde{M})$ be a lift of p and v (v can be lifted because π is a local isometry). Since $\widetilde{M}$ is complete, there exists a geodesic $\tilde{\gamma}$ with $\tilde{\gamma}(0) = \tilde{p}$ and $\tilde{\gamma}'(0) = \tilde{v}$ defined on $\mathbb{R}$. Since π is a local isometry, $\gamma = \pi \circ \tilde{\gamma}$ is a geodesic with $\gamma(0) = p$ and $\gamma'(0) = v$, which implies that M is complete.

Then suppose M is complete. For any $\tilde{p} \in \widetilde{M}$ and $\tilde{v} \in T_{\tilde{p}}(\widetilde{M})$, let $v = \pi(\tilde{v})$ and $v = d\pi_{\tilde{p}}(\tilde{v})$. Since M is complete, there exists a geodesic γ with $\gamma(0) = p$ and $\gamma'(0) = v$ defined on $\mathbb{R}$. By the path lifting property, γ can be lifted to $\tilde{\gamma}$ with $\tilde{\gamma}(0) = \tilde{p}$ and $\tilde{\gamma}'(0) = \tilde{v}$. Hence $\widetilde{M}$ is complete.

- (b) The conclusion is false. For example, we can let $i : \mathbb{R}^2 \setminus \{(0, 0)\} \rightarrow \mathbb{R}^2$ be the inclusion and both manifolds are equipped with the standard Euclidean metric. Then i is a local isometry, and $\mathbb{R}^2$ is complete, but $\mathbb{R}^2 \setminus \{(0, 0)\}$ is not complete.

□

Exercise 2.1.4 ([4] 24 SF T6). Let M be a closed oriented Riemannian manifold, where g_t is a family of smooth Riemannian metrics smoothly depending on $t \in (-\epsilon, \epsilon)$. Suppose there exists a family of eigenfunctions f_t and eigenvalues λ_t smoothly depending on t such that

$$\Delta_{g_t} f_t = \lambda_t f_t,$$

where Δ_{g_t} is the Laplace-Beltrami operator defined using the Riemannian metric g_t . Additionally, assume that f_0 is not a constant function. We define $\dot{\lambda} := \frac{d}{dt}|_{t=0} \lambda_t$ and $\dot{\Delta} := \frac{d}{dt}|_{t=0} \Delta_{g_t}$. Prove the following:

- (1) As λ_0 is an eigenvalue of Δ_{g_0} , let $V_{\lambda_0} := \text{Ker}(\Delta_{g_0} - \lambda_0)$ be the eigenspace of λ_0 , and let $\Pi : L^2(M, g_0) \rightarrow V_{\lambda_0}$ be the orthogonal projection onto the eigenspace. Prove that $\dot{\lambda}$ is an eigenvalue of the operator $\Pi \circ \dot{\Delta} : V_{\lambda_0} \rightarrow V_{\lambda_0}$.
- (2) Let $\varphi_t : M \rightarrow M$ be a 1-parameter family of diffeomorphisms of M and assume $g_t = \varphi_t^* g_0$. Prove that $\dot{\lambda} = 0$.

Proof. (1) We prove that f_0 is an eigenvector of $\Pi \circ \dot{\Delta}$. That is,

$$(\Pi \circ \dot{\Delta} - \dot{\lambda})f_0 = 0$$

which is equivalent to say $\dot{\Delta}f_0 - \dot{\lambda}f_0$ is perpendicular to V_{λ_0} . Differentiating

$$\Delta_{g_t} f_t = \lambda_t f_t$$

we get

$$\dot{\Delta}f_0 + \Delta_{g_0}\dot{f} = \dot{\lambda}f_0 + \lambda_0\dot{f}$$

Then

$$\dot{\Delta}f_0 - \dot{\lambda}f_0 = -(\Delta_{g_0} - \lambda_0)\dot{f}$$

Hence for any $u \in V_{\lambda_0}$,

$$\langle \dot{\Delta}f_0 - \dot{\lambda}f_0, u \rangle = \langle -(\Delta_{g_0} - \lambda_0)\dot{f}, u \rangle = -\langle \dot{f}, (\Delta_{g_0} - \lambda_0)u \rangle = 0$$

where we use that Δ is a self-adjoint operator.

- (2) If $g_t = \varphi_t^* g_0$, then

$$\Delta_{g_t}(f_0 \circ \varphi_t) = \Delta_{g_0}(f_0) \circ \varphi_t = \lambda_0 f_0 \circ \varphi_t$$

Hence $f_0 \circ \varphi_t$ is an eigenvector of Δ_{g_t} with respect to the eigenvalue λ_t . Converse is the same. Hence the spectrum of Δ_{g_0} and Δ_{g_t} are the same. Since M is closed, the spectrum of Laplace operator is discrete. Thus $\lambda_t \equiv \lambda_0$. That is, $\dot{\lambda} = 0$.

□

Chapter 3

Curvature

3.1 Exercise

Exercise 3.1.1 ([3] 26 spring T4). Let (M^3, g) be a Riemannian 3-manifold and fix $p \in M$. Choose an orthonormal basis $\{e_1, e_2, e_3\}$ of $T_p M$ that diagonalizes $Ric_p : T_p M \rightarrow T_p M$ with eigenvalues $\lambda_1, \lambda_2, \lambda_3$.

(a) Show that $\sec_p > 0$ if and only if $\lambda_i + \lambda_j > \lambda_k$ for all distinct i, j, k . In particular, $Ric_p > 0$ (positive definite) does not imply $\sec_p > 0$.

(b) Does $Ric_p = 0$ imply $\sec_p = 0$ in dimension 3? Justify your answer.

Proof. Note that in dimension 3, Ric determines Rm :

$$Rm = (Ric - \frac{S}{4}g) \otimes g$$

Hence if $Ric = 0$, then $S = 0$, thus $Ric = 0$, $\sec = 0$.

For (a), we know that $S = \lambda_1 + \lambda_2 + \lambda_3$. Hence if $\sec_p > 0$, then

$$\begin{aligned} 0 &< \sec_p(e_1, e_2) \\ &= Rm(e_1, e_2, e_2, e_1) \\ &= Ric \otimes g(e_1, e_2, e_2, e_1) - \frac{S}{4}g \otimes g(e_1, e_2, e_2, e_1) \\ &= \lambda_1 + \lambda_2 - \frac{\lambda_1 + \lambda_2 + \lambda_3}{2} \\ &= \frac{1}{2}(\lambda_1 + \lambda_2 - \lambda_3) \end{aligned}$$

Similarly for the other two inequalities. Conversely, if $\lambda_i + \lambda_j > \lambda_k$ for distinct i, j, k , then previous discussion shows that

$$\sec_p(e_1, e_2) > 0, \sec_p(e_2, e_3) > 0, \sec_p(e_1, e_3) > 0$$

Then for general orthonormal $X = X^i e_i, Y = Y^j e_j$, we have

$$\sec_p(X, Y) = \sec_p(e_1, e_2)(X^1 Y^2 - X^2 Y^1)^2 + \sec_p(e_2, e_3)(X^2 Y^3 - X^3 Y^2)^2 + \sec_p(e_1, e_3)(X^1 Y^3 - X^3 Y^1)^2 > 0$$

Thus $\sec_p > 0$. □

Exercise 3.1.2 ([3] 25 autumn T2). Let M be a Riemannian manifold and $\mathcal{K}$ be a set of isometries of M . Suppose that F is a nonempty subset of M which is fixed by every element in $\mathcal{K}$. Show that each connected component of F is a closed totally geodesic submanifold of M .

Proof. To state the proof conveniently, we may assume $\mathcal{K}$ consists of a single isometry ϕ . For any $p \in F$, then $\phi(p) = p$, and there exists a normal neighborhood U of p where $\exp_p$ is a diffeomorphism

$$\exp_p : B_\varepsilon \subseteq T_p M \longrightarrow U \subseteq M$$

Let

$$V = \{v \in T_p M \mid d\phi_p(v) = v\}$$

we claim that

$$F \cap U = \exp_p(B_\varepsilon \cap V)$$

This is because for any $q \in U$, we can write $q = \exp_p(v)$ for some $v \in T_p M$. Let γ be the unique geodesic starting at p with $\gamma'(0) = v$, then $\gamma(1) = q$. Since ϕ is an isometry, $\phi \circ \gamma$ is the unique geodesic starting at $\phi(p) = p$ with $(\phi \circ \gamma)' = d\phi_p(v)$. Hence $\phi(q) = q$ if and only if $d\phi_p(v) = v$.

Since $\exp_p$ is a diffeomorphism and V is a linear subspace of $T_p M$, F is a submanifold of dimension $\dim V$.

F is closed since ϕ is continuous.

To prove that F is totally geodesic, let $p \in F$ and $v \in T_p F$, then $d\phi_p(v) = v$ by previous discussion. Hence the M -geodesic starting at p with initial velocity v will stay in F in some interval $(-\delta, \delta)$. Hence F is totally geodesic. $\square$

Exercise 3.1.3. Let $\mathbb{H}^n = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid x_n > 0\}$ be the upper half-space with the metric defined by $ds^2 = \frac{1}{x_n^2}(dx_1^2 + \dots + dx_n^2)$. Compute the curvature of this manifold.

Proof. The Riemannian metric is given by $g_{ij} = \frac{1}{x_n^2} \delta_{ij}$. By the formula

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$$

we have

$$\Gamma_{ii}^n = \frac{1}{x_n}, \Gamma_{nn}^n = -\frac{1}{x_n}, \Gamma_{ni}^i = \Gamma_{in}^n = -\frac{1}{x_n}, \quad i \neq n$$

By the formula

$$R_{ijk}^l = \partial_i \Gamma_{jk}^l - \partial_j \Gamma_{ik}^l + \Gamma_{im}^l \Gamma_{jk}^m - \Gamma_{jm}^l \Gamma_{ik}^m$$

we have

$$R_{jii}^j = -\frac{1}{x_n^2}, \quad i \neq j$$

Hence

$$Rm = -\frac{1}{2} g \otimes g$$

which implies that $\mathbb{H}^n$ has constant sectional curvature -1 . $\square$

Exercise 3.1.4 ([3] 25 autumn T4). Let $\mathbb{H}^n = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid x_n > 0\}$ be the upper half-space with the metric defined by $ds^2 = \frac{1}{x_n^2}(dx_1^2 + \dots + dx_n^2)$. Let Σ be a k -dimensional minimal submanifold of $\mathbb{H}^n$, where $k < n$. Show that $\text{Ric}(X) \leq 0$, i.e., for every vector field X on Σ , we have $\text{Ric}(X, X) \leq 0$.

Proof. Fix a point $p \in \Sigma$, we may assume $X_p \neq 0$, Normalize X such that $|X_p| = 1$. We may choose a local orthonormal basis $\{e_1, \dots, e_k\}$ at p such that $e_1 = X_p$. Thus we only need to prove that $\text{Ric}(e_1, e_1) = 0$.

By Gauss equation,

$$\begin{aligned} \text{Ric}(e_1, e_1) &= \sum_{i=1}^k Rm(e_i, e_1, e_1, e_i) \\ &= \sum_{i=1}^k \left(\widetilde{Rm}(e_i, e_1, e_1, e_i) + \langle \Pi(e_i, e_i), \Pi(e_1, e_1) \rangle - \langle \Pi(e_i, e_1), \Pi(e_1, e_i) \rangle \right) \\ &= \sum_{i=1}^k \left(\widetilde{Rm}(e_i, e_1, e_1, e_i) - |\Pi(e_i, e_1)|^2 \right) + \left\langle \sum_{i=1}^k \Pi(e_i, e_i), \Pi(e_1, e_1) \right\rangle \end{aligned}$$

where $\widetilde{Rm}$ denotes the curvature of $\mathbb{H}^n$. Since $\mathbb{H}^n$ has negative constant sectional curvature,

$$\widetilde{Rm}(e_i, e_1, e_1, e_i) = \sec(e_i, e_1) \leq 0, \quad \forall i$$

Since Σ is a minimal submanifold,

$$0 = \text{tr}_g \Pi = \sum_{i=1}^k \Pi(e_i, e_i)$$

Thus we conclude that $\text{Ric}(e_1, e_1) \leq 0$. □

Exercise 3.1.5 ([3] 25 spring T2). Let $B^n(r) = \{x \in \mathbb{R}^n \mid |x| < r\}$. Compute the sectional curvature of

$$g = \frac{r^4}{(r^2 - |x|^2)^2} \sum_{i=1}^n dx^i \otimes dx^i$$

Proof. The Riemannian metric is given by $\frac{r^4}{(r^2 - |x|^2)^2} \delta_{ij}$. By the formula

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$$

we have

$$\Gamma_{ii}^i = \frac{2x^i}{r^2 - |x|^2}, \quad \Gamma_{jj}^i = -\frac{2x^i}{r^2 - |x|^2}, \quad \Gamma_{ji}^i = \Gamma_{ij}^j = \frac{2x^i}{r^2 - |x|^2}, \quad i \neq j$$

By the formula

$$R_{ijk}^l = \partial_i \Gamma_{jk}^l - \partial_j \Gamma_{ik}^l + \Gamma_{im}^l \Gamma_{jk}^m - \Gamma_{jm}^l \Gamma_{ik}^m$$

we have

$$R_{jii}^j = -\frac{4r^2}{(r^2 - |x|^2)^2}, \quad i \neq j$$

Hence

$$Rm = -\frac{2}{r^2} g \otimes g$$

which implies that $(B^n(r), g)$ has constant sectional curvature $-\frac{4}{r^2}$. □

Exercise 3.1.6. Recall that a Riemannian metric h on a Lie group G is said to be **left-invariant** if

$$L_g^* h = h$$

for all $g \in G$. A **right-invariant metric** is defined similarly. A metric that is both left-invariant and right-invariant is said to be **bi-invariant**. Prove that any compact Lie group admits a bi-invariant metric.

Proof. First take ω to be a right-invariant volume form, and $\langle -, - \rangle$ be a right-invariant metric. Define

$$h(X, Y) = \int_G \langle dL_g(X), dL_g(Y) \rangle \omega$$

Then h is left-invariant:

$$\begin{aligned} L_a^* h(X, Y) &= h(dL_a(X), dL_a(Y)) \\ &= \int_G \langle dL_g(dL_a(X)), dL_g(dL_a(Y)) \rangle \omega \\ &= \int_G \langle dL_g(X), dL_g(Y) \rangle \omega \\ &= h(X, Y) \end{aligned}$$

where the third equality is due to integration, and right-invariant:

$$\begin{aligned}
R_a^* h(X, Y) &= h(dR_a(X), dR_a(Y)) \\
&= \int_G \langle dL_g(dR_a(X)), dL_g(dR_a(Y)) \rangle \omega \\
&= \int_G \langle dR_a(dL_g(X)), dR_a(dL_g(Y)) \rangle \omega \\
&= \int_G \langle dL_g(X), dL_g(Y) \rangle \omega \\
&= h(X, Y)
\end{aligned}$$

where the fourth equality uses right-invariance of $\langle -, - \rangle$. Thus h satisfies the condition. $\square$

Exercise 3.1.7 ([3] 25 spring T6). Let (G, h) be a Lie group with a bi-invariant metric h .

(1) Show that for all left-invariant vector fields X, Y, Z ,

$$h([X, Y], Z) = h(X, [Y, Z])$$

(2) Show that for all left-invariant vector fields X, Y ,

$$\nabla_X Y = \frac{1}{2}[X, Y]$$

where ∇ is the Levi-Civita connection.

Proof. Recall the adjoint representation of a Lie group.

(1) Since h is bi-invariant, it is invariant under adjoint representation. Hence

$$h(\text{Ad}(\exp(tY))X, \text{Ad}(\exp(tY))Z) = h(X, Z), \quad \forall t, X, Y, Z$$

Differential this equation at $t = 0$, we get

$$\begin{aligned}
0 &= \left. \frac{d}{dt} \right|_{t=0} h(\text{Ad}(\exp(tY))X, \text{Ad}(\exp(tY))Z) \\
&= h\left(\left. \frac{d}{dt} \right|_{t=0} \text{Ad}(\exp(tY))X, \text{Ad}(\exp(tY))Z \right|_{t=0} + h(\text{Ad}(\exp(tY))X \Big|_{t=0}, \left. \frac{d}{dt} \right|_{t=0} \text{Ad}(\exp(tY))Z) \\
&= h(\text{ad}_Y(X), Z) + h(X, \text{ad}_Y(Z)) \\
&= h([Y, X], Z) + h(X, [Y, Z])
\end{aligned}$$

Thus

$$h([X, Y], Z) = h(X, [Y, Z])$$

(2) By the formula of Levi-Civita connection,

$$\langle \nabla_X Y, Z \rangle = \frac{1}{2} \left(X \langle Y, Z \rangle + Y \langle Z, X \rangle - Z \langle X, Y \rangle - \langle X, [Y, Z] \rangle - \langle Y, [X, Z] \rangle + \langle Z, [X, Y] \rangle \right)$$

Note that $\langle X, Y \rangle$ is always constant for left-invariant vector fields X, Y . Hence the first three terms vanish.

And by (1),

$$\langle X, [Y, Z] \rangle + \langle Y, [X, Z] \rangle = 0$$

Hence

$$\langle \nabla_X Y, Z \rangle = \frac{1}{2} \langle [X, Y], Z \rangle$$

for all Z . Thus

$$\nabla_X Y = \frac{1}{2}[X, Y]$$

□

Exercise 3.1.8. Let (G, h) be a Lie group with a bi-invariant metric h . Compute the curvature of (G, h) .

Proof. Using last exercise, we have

$$\begin{aligned} R(X, Y)Z &= \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z \\ &= \frac{1}{4}[X, [Y, Z]] - \frac{1}{4}[Y, [X, Z]] - \frac{1}{2}[[X, Y], Z] \\ &= \frac{1}{4}[Z, [X, Y]] \end{aligned}$$

In particular, for X, Y orthonormal,

$$\begin{aligned} \sec(X, Y) &= Rm(X, Y, Y, X) \\ &= \langle R(X, Y)Y, X \rangle \\ &= \frac{1}{4}\langle [Y, [X, Y]], X \rangle \\ &= \frac{1}{4}|[X, Y]|^2 \end{aligned}$$

We find that $\sec \leq 0$.

□

Exercise 3.1.9 ([3] 24 autumn T3). Let (M, g) be a connected Riemannian manifold of dimension n . Suppose that there exists some $f \in C^\infty(M, \mathbb{R})$ such that

$$\text{Ric}(g) = (n - 1)fg$$

- (1) If $n = 2$, is f necessary a constant?
- (2) If $n = 2024$, is f necessary a constant?

Proof. This is just Schur's lemma.

- (1) No. In dimension 2, $\text{Ric} = Kg$, hence we may take a surface whose Gaussian is not constant.
- (2) Yes. We prove that if $n \leq 3$, $\text{Ric} = fg$, then f is constant. Write in local coordinates:

$$R_{ij} = fg_{ij}, \quad R_i^j = f\delta_i^j$$

Hence the scalar curvature is

$$S = R_i^i = nf$$

Plug into the contracted Bianchi identity

$$\nabla_j R_i^j = \frac{1}{2}\nabla_i S$$

We get

$$\nabla_j f = \frac{n}{2}\nabla_j f$$

Since $n \leq 3$, we have $\nabla_j f = 0$ for all j . Thus f is a constant.

□

Exercise 3.1.10 ([4] 25SF T2). The Lie group $SU(2)$ is diffeomorphic to the 3-sphere S^3 , with Lie algebra spanned by

$$X_1 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad X_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad X_3 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

A positive-definite inner product on the Lie algebra determines a left-invariant Riemannian metric on the Lie group. The Berger metrics g_ϵ on S^3 are obtained by making the inner product on the Lie algebra such that $\epsilon^{-1}X_1, X_2$, and X_3 form an orthonormal frame, where $\epsilon > 0$ is a positive number.

(1) Let ∇ be the Levi-Civita connection for g_ϵ . Prove that $\nabla_{X_i} X_i = 0$ for $i = 1, 2, 3$.

(2) Compute the scalar curvature for g_ϵ .

Proof. First note that we have the relation $[X_i, X_j] = \varepsilon_{ijk} X_k$.

(1) By the formula of Levi-Civita connection,

$$\langle \nabla_X Y, Z \rangle = \frac{1}{2} \left(X \langle Y, Z \rangle + Y \langle Z, X \rangle - Z \langle X, Y \rangle - \langle X, [Y, Z] \rangle - \langle Y, [X, Z] \rangle + \langle Z, [X, Y] \rangle \right)$$

Note that $\langle X, Y \rangle$ is always constant for left-invariant vector fields X, Y . Hence the first three terms vanish. Thus

$$\langle \nabla_{X_i} X_i, Z \rangle = \frac{1}{2} \left(-\langle X_i, [X_i, Z] \rangle - \langle X_i, [X_i, Z] \rangle + \langle Z, [X_i, X_i] \rangle \right) = 0$$

for all Z . Hence $\nabla_{X_i} X_i = 0$.

(b) Denote the orthonormal basis by $e_1 = \varepsilon^{-1} X_1, e_2 = X_2, e_3 = X_3$. Then direct calculation shows that

$$\Gamma_{12}^3 = \frac{2}{\varepsilon} - \varepsilon, \Gamma_{23}^1 = \varepsilon, \Gamma_{31}^2 = \varepsilon, \Gamma_{21}^3 = -\varepsilon, \Gamma_{32}^1 = -\varepsilon, \Gamma_{13}^2 = \varepsilon - \frac{2}{\varepsilon}$$

By the formula

$$R_{ijk}^l = \partial_i \Gamma_{jk}^l - \partial_j \Gamma_{ik}^l + \Gamma_{im}^l \Gamma_{jk}^m - \Gamma_{jm}^l \Gamma_{ik}^m$$

we get

$$R_{ijj}^i = -\Gamma_{jm}^i \Gamma_{ij}^m$$

Hence

$$S = R_{ij}^{ji} = R_{ijj}^i = -\varepsilon^2 - 2\varepsilon\left(\frac{2}{\varepsilon} - \varepsilon\right) - \varepsilon^2 + 2\varepsilon\left(\varepsilon - \frac{2}{\varepsilon}\right) = 2\varepsilon^2 - 8$$

□

Exercise 3.1.11 ([4] 25 SF T5). Let (M, g) be a closed, oriented, two-dimensional smooth Riemannian manifold with Gaussian curvature K . Suppose u is a smooth function, and define a new metric $\tilde{g} := e^{2u}g$ in the same conformal class as g , with Gaussian curvature $\tilde{K}$.

(1) Prove that u satisfies the equation

$$\Delta u - K + \tilde{K}e^{2u} = 0,$$

where Δ is the Laplace-Beltrami operator defined by the metric g .

(2) Suppose the Euler characteristic $\chi(M) = 0$. Prove that either $\tilde{K} \equiv 0$ or

$$\int_M \tilde{K} e^{2f} d\text{Vol}_g < 0,$$

where f is a solution to the equation $\Delta f = K$.

Proof. (1) In general, we have

$$\tilde{S} = e^{-2u}(S - 2(n-1)\Delta u - (n-1)(n-2)|\nabla u|^2)$$

When $n = 2$, this gives

$$\tilde{S} = e^{-2u}(S - 2\Delta u)$$

Since $S = 2K$, we have

$$\Delta u - K + \tilde{K}e^{2u} = 0$$

(2) If $K = \Delta f$, then

$$\tilde{K} = e^{-2u} \Delta(f - u)$$

Hence by integration by parts,

$$\begin{aligned} \int_M \tilde{K} e^{2f} dV &= \int_M e^{2(f-u)} \Delta(f - u) \\ &= \int_{\partial M} e^{2(f-u)} \langle \nabla(f - u), N \rangle dV_{\partial M} - \int_M \langle \nabla(e^{2(f-u)}), \nabla(f - u) \rangle dV \\ &= -2e^{2(f-u)} \int_M |\nabla(f - u)|^2 dV \\ &\leq 0 \end{aligned}$$

and the equality holds if and only if $\nabla(f - u) \equiv 0$. Hence $\tilde{K} \equiv 0$. □

Exercise 3.1.12 ([4] 24 SF T4). *Let (M, g) be a closed oriented n -dimensional Riemannian manifold. Prove that the scalar curvature $S(p)$ at $p \in M$ is given by*

$$S(p) = \frac{n}{\omega_{n-1}} \int_{S^{n-1}} Ric_p(V, V) dS^{n-1},$$

where ω_{n-1} is the area of the unit sphere S^{n-1} in $T_p M$, $V \in S^{n-1}$ are unit vector fields, and dS^{n-1} is the area element on S^{n-1} .

Proof. Choose an orthonormal basis $\{e_1, \dots, e_n\}$ of $T_p M$. Then

$$\begin{aligned} \frac{n}{\omega_{n-1}} \int_{S^{n-1}} Ric_p(V, V) dS^{n-1} &= \frac{n}{\omega_{n-1}} \int_{S^{n-1}} Ric_p(x^i e_i, x^j e_j) dS^{n-1} \\ &= \frac{n}{\omega_{n-1}} Ric_p(e_i, e_j) \int_{S^{n-1}} x^i x^j dS^{n-1} \end{aligned}$$

By symmetric, if $i \neq j$,

$$\int_{S^{n-1}} x^i x^j dS^{n-1} = 0$$

Moreover,

$$\int_{S^{n-1}} (x^1)^2 dS^{n-1} = \dots = \int_{S^{n-1}} (x^n)^2 dS^{n-1} = \frac{1}{n} \int_{S^{n-1}} ((x^1)^2 + \dots + (x^n)^2) dS^{n-1} = \frac{\omega_{n-1}}{n}$$

Thus

$$\frac{n}{\omega_{n-1}} \int_{S^{n-1}} Ric_p(V, V) dS^{n-1} = \sum_i Ric_p(e_i, e_i) = S(p)$$

□

Chapter 4

Jacobi fields, topology

4.1 Theorems

Theorem 4.1.1 (Rauch comparison). *Suppose J_i are Jacobi fields of manifolds with curvatures R_i . Suppose that $\lambda_-(R_1) \geq \lambda_+(R_2)$, and one of the following two condition holds:*

$$(1) \ J_i(0) = 0, |J'_1(0)| = |J'_2(0)|.$$

$$(2) \ J'_i(0) = 0, |J_1(0)| = |J_2(0)|.$$

Then $|J_1| \leq |J_2|$ up to the first zero of J_1 .

Theorem 4.1.2 (Bishop-Gromov). *Suppose (M, g) is a n -dimensional complete Riemannian manifold, and $\text{Ric} \geq (n-1)k$. Denote S_n^k by the complete, simply-connected space with constant sectional curvature k . Fix a point $p \in M$, then*

$$f(r) = \frac{\text{vol}(B_r(p))}{\text{vol } B_r}$$

is decreasing and $\lim_{r \rightarrow 0^+} f(r) = 1$. In particular,

$$\text{vol}(B_r(p)) \leq \text{vol } B_r$$

Theorem 4.1.3 (Cartan-Hadamard). *Suppose M is a complete, simply-connected Riemannian manifold with $\text{sec} \leq 0$, then M is diffeomorphic to $\mathbb{R}^n$ and $\exp_p : T_p M \rightarrow M$ gives a diffeomorphism.*

Theorem 4.1.4 (Killing-Hopf). *Suppose M is a complete Riemannian manifold with constant sectional curvature, then the universal cover $\tilde{M}$ if M is isometric to*

$$\begin{cases} S^n(R), & \text{sec} > 0 \\ \mathbb{R}^n, & \text{sec} = 0 \\ \mathbb{H}^n(R), & \text{sec} < 0 \end{cases}$$

Theorem 4.1.5 (Myers). *Suppose (M, g) is a n -dimensional complete Riemannian manifold, and there exists a constant $R > 0$ such that $\text{Ric} \geq \frac{n-1}{R^2}$, then M is compact with diameter $\leq \pi R$, and $\pi_1(M)$ is finite.*

Theorem 4.1.6 (Cheng). *Assumption as above, suppose $\text{diam}(M) = \pi R$, then (M, g) is isometric to $S^n(R)$.*

Theorem 4.1.7. *Suppose M is a closed Riemannian manifold with $\text{Ric} > 0$, then the first Betti number b_1 of M is 0. If $\text{Ric} \leq 0$, then $b_1 \leq n$.*

Theorem 4.1.8 (Bochner formula). *Suppose V is a vector field, $A = \nabla_- V$ is a operator, $W = \nabla_V V$. Then*

$$\nabla_V \text{div } V + \text{tr}(A^2) + \text{Ric}(V) = \text{div } W$$

In particular, if V is a gradient field $V = \nabla f$, then

$$\langle \nabla f, \nabla \Delta f \rangle + |\nabla^2 f|^2 + \text{Ric}(\nabla f) = \frac{1}{2} \Delta |\nabla f|^2$$

Theorem 4.1.9 (Preissman). *Suppose M is a compact Riemannian manifold with $\text{sec} < 0$, then any nontrivial abelian subgroup of $\pi_1(M)$ is isomorphic to $\mathbb{Z}$.*

Theorem 4.1.10. *Suppose M is a compact Riemannian manifold with $\text{sec} < 0$, then $\pi_1(M)$ is not abelian.*

Theorem 4.1.11 (Synge-Weinstein). *Suppose M is a compact Riemannian manifold with $\text{sec} > 0$. For $\dim M$ even, if M is orientable, then $\pi_1(M) = 0$; if M is nonorientable, then $\pi_1(M) \cong \mathbb{Z}_2$. For $\dim M$ odd, M must be orientable.*

4.2 Exercise

Exercise 4.2.1 (Berger comparison, [3] 25 autumn T8). *Let M be a complete Riemannian manifold with nonnegative sectional curvature and $\gamma : [0, 1] \rightarrow M$ be a minimizing geodesic with $\gamma(0) = p, \gamma(1) = q$. Suppose that X is a parallel vector field along γ such that $X \perp \gamma'$.*

(a) *Show that for sufficiently small $t > 0$, $\text{dist}(\exp_p(tX(0)), \exp_q(tX(1))) \leq \text{dist}(p, q)$.*

(b) *State the condition for which the equality in (a) holds (no proof needed).*

Proof. (a) Let F be a variation defined by

$$\begin{aligned} F : (-\varepsilon, \varepsilon) \times [0, 1] &\longrightarrow M \\ (s, t) &\longmapsto \exp_{\gamma(t)}(sX(t)) \end{aligned}$$

where $\exp_{\gamma(t)}$ is a diffeomorphism. Then for fixed $t_0 \in (0, 1)$, $F(-, t_0)$ is a geodesic, and we can view F as a geodesic variation of $F(-, t_0)$ where the variation field is $J(s) = \frac{\partial F}{\partial t}(s, t_0)$ and

$$J(0) = \frac{\partial F}{\partial t}(0, t_0) = \gamma'(t_0)$$

We consider a Jacobi field $\bar{J}(s)$ in Euclidean space with $|\bar{J}(0)| = |\gamma'(t_0)|$. Since $\bar{J}(s)$ is constant, $|\bar{J}(s)| = |\gamma'(t_0)|$ for any s . Now since M has nonnegative sectional curvature, by Rauch comparison theorem, we have

$$|J(s)| \leq |J'(s)| = |\gamma'(t_0)|, \forall s$$

Thus

$$\begin{aligned} \text{dist}(\exp_p(sX(0)), \exp_q(sX(1))) &= \text{dist}(F(s, 0), F(s, 1)) \\ &\leq \int_0^1 \left| \frac{\partial F}{\partial t}(s, t) \right| dt \\ &\leq \int_0^1 |\gamma'(t)| dt \\ &= \text{dist}(p, q) \end{aligned}$$

(b) If the equality in (a) holds for some t , then by equality condition in Rauch comparison theorem, M has sectional curvature 0 in $F([0, t] \times [0, 1])$, and $J(s)$ is parallel along $F(-, t_0)$ for any $t_0 \leq t$. Hence we conclude that $F([0, t] \times [0, 1])$ is a flat totally geodesic rectangle. □

Exercise 4.2.2 ([3] 26 spring T2). Consider $\mathbb{RP}^n$, $n \geq 3$, with the standard metric induced from the standard sphere S^n of sectional curvature 1 via the antipodal quotient. Fix a point $[p] \in \mathbb{RP}^n$, where $p \in S^n$. Describe the cut locus $\text{Cut}([p]) \subset \mathbb{RP}^n$ explicitly.

Proof. Recall that the cut locus is the set of points that fail to be the shortest geodesics. Hence

$$\text{Cut}([p]) = \{[q] \in \mathbb{RP}^n \mid p \perp q\}$$

is a submanifold $\simeq \mathbb{RP}^{n-1}$. □

Exercise 4.2.3 ([3] 25 spring T10). Let (M, g) be an n -dimensional compact oriented Riemannian manifold with positive sectional curvature. Given an isometry $F : M \rightarrow M$ such that F preserves the orientation when n is even, F changes the orientation when n is odd. Show that F has a fixed point.

Proof. Suppose, for the contrary, that F doesn't have a fixed point. Then we can choose a point $p_0 \in M$ such that $d(p_0, F(p_0)) > 0$ is minimal. Let $\gamma : [0, 1] \rightarrow M$ be a minimal geodesic connecting p_0 and $F(p_0)$. Since F is an isometry, $F(\gamma)$ is a geodesic connecting $F(p_0)$ and $F^2(p_0)$.

For any $c \in \gamma$, we have

$$\begin{aligned} d(c, F(c)) &\leq d(c, F(p_0)) + d(F(p_0), F(c)) \\ &= d(c, F(p_0)) + d(p_0, c) \\ &= d(p_0, F(p_0)) \end{aligned}$$

But by our choice of p_0 , we should have $d(c, F(c)) \geq d(p_0, F(p_0))$. Thus the above equality holds, which implies that $c \rightarrow F(p_0) \rightarrow F(c)$ is a minimal geodesic. Hence we have

$$dF_p(\gamma'(0)) = \gamma'(1)$$

Denote the linear space consisting of all parallel vector fields along γ starting at p_0 ending at $F(p_0)$ by V . Then γ' is an eigenvector of dF_p . Hence we have a restricted linear map $dF_p|_{\gamma'^\perp}$. This is a orthogonal transformation since F is an isometry. By assumption, it preserves the orientation when n is even, and changes the orientation when n is odd. By considering determinant, we know that $dF_p|_{\gamma'^\perp}$ has an eigenvalue of 1. That is, there exists a parallel normal vector field N along γ such that $F(N(0)) = N(1)$.

Now consider a variation $G(s, t) = \exp_{\gamma(t)}(sN(t))$ along γ . Then $F(G(s, 0)) = G(s, 1)$ since N is parallel and F is an isometry. By Berger comparison theorem, $d(G(s, 0), G(s, 1)) < d(p_0, F(p_0))$, which is a contradiction. (Or one can use energy variation.) □

Exercise 4.2.4 ([3] 26 spring T6). Let (M, g) be a closed Riemannian manifold and let X be a Killing vector field. Show that if $\text{Ric} < 0$, then $X \equiv 0$.

Proof. By Bochner formula,

$$\nabla_X \text{div} X + \text{tr}(A^2) + \text{Ric}(X) = \text{div}(W)$$

where $A = \nabla_- X$, $W = \nabla_X X$. Since X is a Killing vector field, A is anti-linear, hence

$$\begin{aligned} \text{div}(X) &= \text{tr}(\nabla X) = \text{tr}(A) = 0 \\ \text{tr}(A^2) &= \sum_i \langle A^2 e_i, e_i \rangle = \sum_i -\langle A e_i, A e_i \rangle \leq 0 \end{aligned}$$

Thus we do integration on M for the initial equality and get

$$0 \geq LHS = RHS = 0$$

where we used the divergence theorem. And the equality holds if and only if $X \equiv 0$ since $\text{Ric} < 0$. □

Exercise 4.2.5 ([3] 25 autumn 10). Let (M, g) be a complete compact oriented Riemannian manifold with dimension $n \geq 2$. We say a vector field X on M is a conformal Killing field if $\mathcal{L}_X g = fg$ for some smooth function f , where $\mathcal{L}_X$ denotes the Lie derivative with respect to X . If $\text{Ric}(M) \leq 0$, show that every conformal Killing field on M is parallel. (Hint: Consider $\nabla_X X - \text{div}(X)X$.)

Proof. Suppose X is a conformal Killing field. For any $Y, Z \in \mathfrak{X}(M)$, we have

$$\begin{aligned} f\langle Y, Z \rangle &= \mathcal{L}_X g(Y, Z) \\ &= X\langle Y, Z \rangle - \langle [X, Y], Z \rangle - \langle Y, [X, Z] \rangle \\ &= \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle - \langle \nabla_X Y - \nabla_Y X, Z \rangle - \langle Y, \nabla_X Z - \nabla_Z X \rangle \\ &= \langle \nabla_Y X, Z \rangle + \langle Y, \nabla_Z X \rangle \end{aligned}$$

If we write $A = \nabla X \in \text{End}(\mathfrak{X}(M))$, then the above equality can be written as

$$A + A^T = fI_n$$

By an easy version Bochner formula, we have

$$\nabla_X \text{div} X + \text{tr}(A^2) + \text{Ric}(X) = \text{div} W$$

where $W = \nabla_X X$ (One can derive this from taking trace of the expansion of $\nabla_X A$). Then we have

$$\text{div} X = \text{tr}(\nabla X) = \text{tr}(A) = \frac{1}{2} \text{tr}(A + A^T) = \text{tr}(fI_n) = \frac{nf}{2}$$

and

$$\text{tr}(A^2) = \sum_{i=1}^n \langle A^2 e_i, e_i \rangle = \sum_{i=1}^n \langle (fI_n - A^T) A e_i, e_i \rangle = f \text{tr}(A) - \sum_{i=1}^n \langle A e_i, A e_i \rangle = \frac{nf^2}{2} - |A|^2$$

By expanding div in local coordinates, we have

$$\text{div}(gZ) = g \text{div} Z + Zg, \quad \forall g \in C^\infty(M), Z \in \mathfrak{X}(M)$$

Hence

$$\text{div}(\text{div}(X)X) = \text{div}(X)^2 + \nabla_X \text{div} X = \frac{n^2 f^2}{4} + \nabla_X \text{div} X$$

Finally, we consider

$$\text{div}(\nabla_X X - \text{div}(X)X) = \text{div} W - \text{div}(\text{div}(X)X) = \frac{n(2-n)f^2}{4} + \text{Ric}(X) - |A|^2$$

Integrate both sides on M , and by divergence theorem, we have

$$0 = \int_M \frac{n(2-n)f^2}{4} + \text{Ric}(X) - |A|^2$$

Since $\text{Ric}(M) \leq 0$, $\text{Ric}(X) \leq 0$, thus all three terms in the right side are ≤ 0 . Hence all equal to 0. In particular, $A = 0$, i.e. X is parallel. $\square$

Exercise 4.2.6 ([3] 25 spring T4). Let (M, g) be a Riemannian manifold and X is a Killing field. If γ is a geodesic, show that $J(t) = X \circ \gamma(t)$ is a Jacobi field along γ .

Proof. Direct calculation gives

$$\begin{aligned} \frac{D^2}{dt^2} J + R(J, \gamma')\gamma' &= \nabla_{\gamma'} \nabla_{\gamma'} X + \nabla_X \nabla_{\gamma'} \gamma' - \nabla_{\gamma'} \nabla_X \gamma' - \nabla_{[X, \gamma']} \gamma' \\ &= \nabla_{\gamma'} [\gamma', X] - \nabla_{[X, \gamma']} \gamma' \end{aligned}$$

Now we prove that $[X, \gamma'] = 0$, then J is a Jacobi field. Note that by definition,

$$\mathcal{L}_X g = g$$

Hence if we denote the flow of X by φ_t , then $\phi_t^* g = g$. Thus ϕ_t is an isometry. Since γ is a geodesic, $\phi(\gamma)$ is a geodesic. Hence if we denote the geodesic flow by ψ_s , we have

$$\phi_t \circ \psi_s = \psi_s \circ \phi_t$$

Thus $[X, \gamma'] = 0$. □

Chapter 5

Curves and surfaces

5.1 Exercise

Exercise 5.1.1 ([4] 24 SF T2). *Let $\Sigma \subset \mathbb{R}^3$ be an embedded surface in $\mathbb{R}^3$. Prove that if Σ is closed, then Σ cannot be a minimal surface.*

Proof. Since Σ is closed, we get choose a point $p \in \Sigma$ such that $|OP|$ is maximal. Suppose the principal direction at P is represented by a curve γ such that

$$s(\gamma'(0)) = \kappa\gamma'(0)$$

where s is the shape operator. Then we have

$$0 = \frac{d}{dt} \Big|_{t=0} |\gamma(t)|^2 = 2\langle \gamma'(0), \gamma(0) \rangle$$

Hence the normal vector is $N = \frac{\gamma(0)}{|\gamma(0)|}$. Moreover,

$$0 > \frac{d}{dt} \Big|_{t=0} |\gamma(t)|^2 = 2\langle \gamma''(0), \gamma(0) \rangle + 2|\gamma'(0)|^2$$

Thus

$$\begin{aligned} \kappa|\gamma'(0)|^2 &= \langle s(\gamma'(0)), \gamma'(0) \rangle \\ &= \Pi_N(\gamma'(0), \gamma'(0)) \\ &= \langle \gamma''(0), \frac{\gamma(0)}{|\gamma(0)|} \rangle \\ &< -|\gamma'(0)|^2 \end{aligned}$$

gives

$$\kappa < -\frac{1}{|\gamma'(0)|} < 0$$

Hence

$$H(P) = \kappa_1 + \kappa_2 < 0$$

which implies that Γ is not a minimal surface. □

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